Machine Learning-Based Prediction of Next-Hour Urban PM2.5 Concentration Using Meteorological and Atmospheric Pollutant Data

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***Abstract*—Air pollution, particularly fine particulate matter (PM2.5) has become a significant environmental concern due to its effects on air quality and human well-being. The concentration of PM2.5 changes over time and is influenced by other air pollutants, meteorological conditions, temporal patterns, and geographical location, making accurate short-term forecasting an important research problem. This study focuses on developing a machine-learning-based framework for one-hour-ahead PM2.5 forecasting using the Beijing Multi-Site Air Quality Dataset which contains hourly air-quality and meteorological observations collected from 12 monitoring stations between 2013 and 2017. The study investigates whether commonly used machine-learning models can effectively predict the PM2.5 concentration of the following hour using information available at the current hour and whether increasing model complexity necessarily improves short-term forecasting performance. The methodology includes data integration and pre-processing, exploratory data analysis, feature preparation, construction of the next-hour PM2.5 prediction target and chronological separation of training and testing observations. A persistence baseline is established as a simple forecasting benchmark while Linear Regression, Decision Tree, Random Forest, and XGBoost are employed as the machine-learning models. Their predictive performances are assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). In addition, an expanding-window rolling-validation strategy is applied to examine model performance across different future periods. Through this framework, the study aims to provide a systematic and interpretable comparison of simple and more complex machine-learning approaches for short-term PM2.5 forecasting while emphasizing appropriate time-aware evaluation and baseline comparison.**

***Index Terms*—PM2.5 Forecasting, Air Pollution, Machine Learning, Linear Regression, Time-Series Forecasting, Rolling Validation.**

I. Introduction

Air pollution is an environmental problem of great importance, among which particulate matter with an aerodynamic diameter of less than 2.5 μm (PM2.5) plays an important role in both air quality and human health. Short-term PM2.5 forecast can provide air-quality monitoring and early warning of multi-level pollution [1], [2].

To predict PM2.5, the Random Forest model, XGBoost, LightGBM, LSTM, and hybrid deep-learning models have been studied in recent years [4], [5], [8], [10], [11]. These methods have proven to be effective, but it has been found that more complicated models do not necessarily yield more accurate forecasts for all forecasting horizons and data sets.

This work explores the one-hour-ahead PM2.5 prediction based on the Beijing Multi-Site Air Quality Dataset. Four machine learning models, namely Linear Regression, Decision Tree, Random Forest, and XGBoost models, are tested and compared with a persistence baseline based on the metrics MAE, RMSE, and R^2. A chronological train-test split and rolling validation using a window of increasing size are used to maintain the time order and to ensure model stability. The study focuses on comparing the performance of complex machine-learning models with simpler models in short-term PM2.5 predictions.

II. Related Work

Various statistical, machine-learning and deep-learning techniques have been widely explored for PM2.5 forecasting. Previous pollutant- and meteorological-based studies have shown that pollutant concentrations and meteorological factors can be useful in the estimation and prediction of PM2.5 concentrations [1], [2]. As more and more large-scale air-quality datasets have become available, machine-learning techniques have become more and more common due to their capacity to model complex relationships between environmental variables.

Other studies have focused on prediction of PM2.5 using tree-based and ensemble-learning methods. The application of Random Forest and XGBoost have been used to combine pollutant, meteorological, and other environmental information, showing good prediction abilities [4]. Another line of research has been the inclusion of spatial and temporal features in the development of spatiotemporal XGBoost models, which can be used to enhance the accuracy of PM2.5

prediction and estimation [11]. Likewise, the LightGBM-based models have been studied for the prediction of air quality as an effective gradient-boosting model [8], [14]. The studies show that ensemble methods are suitable for interpretation of the nonlinear relationships in air quality data.

Recent studies have investigated deep-learning models to be used to capture the temporal dependence in PM2.5 observations. To capture the key temporal features and enhance the forecasting accuracy, attention-based LSTM models have been used for identification of temporal features [5]. Hybrid CNN-LSTM architectures have also been studied, where the feature-extraction ability of CNN networks is combined with the temporal modelling ability of LSTM networks [10], [13]. Moreover, approaches based on transformer were added to model complex temporal relationships with attention mechanisms [15].

While models are more complicated in previous studies, they show promise, but their performance is tied to the type of data set, features fed into the model, the prediction horizon, and the evaluation methodology. Thus, in any forecasting context, more complex models cannot be expected to perform better than simpler models. The present study draws inspiration from this observation to compare the performance of Linear Regression, Decision Tree, Random Forest, and XGBoost with a persistence baseline in the evaluation framework of chronological evaluation. The goal is to assess the value added from higher complexity models for one-hour-ahead PM2.5 prediction.

III. Methodology

The proposed framework comprises of data collection and integration, preprocessing, exploratory data analysis, feature engineering, chronological data splitting, model development, and performance evaluation. The Beijing Multi-Site Air Quality Dataset was created for one-hour-ahead PM2.5 prediction, i.e., air quality data from the previous hour were used to predict the concentration of PM2.5 in the next hour. Four machine-learning models were compared to a persistence baseline, and temporal stability assessed by rolling validation.

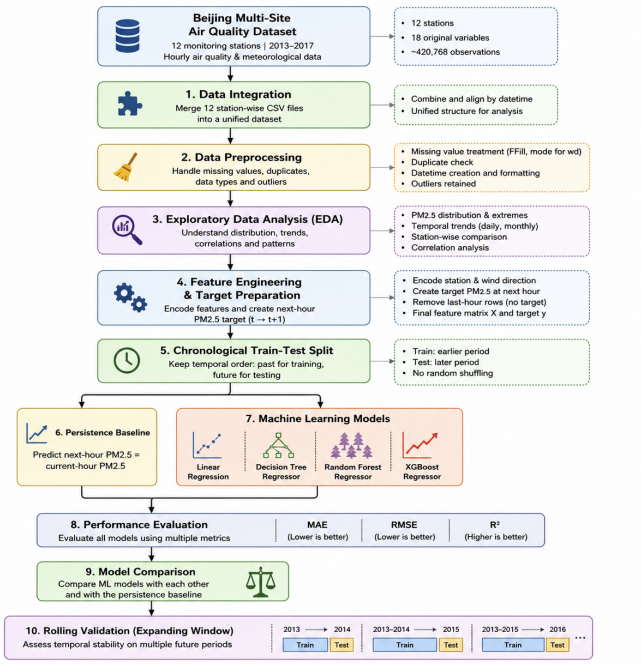


Fig. 1. Overall workflow of the proposed PM2.5 forecasting methodology.

A. Dataset and Data Pre-processing

The Beijing Multi-Site Air Quality dataset consists of hourly air quality and weather observations at twelve air quality monitoring stations in Beijing, China, in the period 2013–2017. The variables contained in this dataset are air pollution indicators: particulate matter (PM2.5, PM10), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), carbon monoxide (CO), ozone (O_3), and weather data: temperature, pressure, dew point, wind direction, and speed.

All the 12 station-wise datasets were merged to form a composite dataset and sorted chronologically. Duplicate observations and outliers were investigated in the pre-processing stage and missing observations were either interpolated or forward filled as appropriate. The year, month, day, and hour information were used to create a `datetime` variable, as this information is temporal and needed for forecasting.

B. Exploratory Data Analysis

Prior to the development of the model, exploratory data analysis (EDA) was carried out to gain insight into the distribution, temporal behaviour, and relations between the variables. A histogram was used to analyse the distribution of PM2.5 concentration and temporal analysis was conducted to identify the difference in the PM2.5 average concentration in different months. Correlation analysis and a few selected scatter plots were performed to further analyze the relationship between PM2.5 and other pollutants as well as meteorological

variables.

The correlation analysis showed that PM2.5 was related to some pollutant and meteorological variables, which suggested that these variables could be incorporated into the forecasting framework. The results from EDA also showed significant differences in the levels of PM2.5 within the period of observation, which highlights the need to account for both pollution and meteorological data for short-term forecasting.

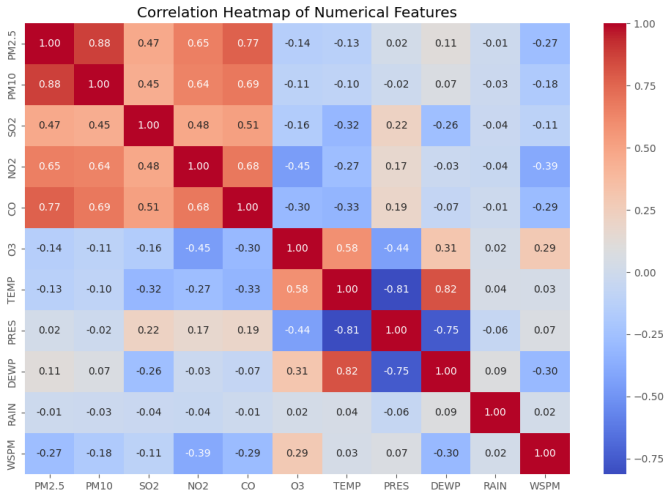


Fig. 2. Correlation Heatmap of Numerical Features.

C. Feature Engineering and Target Preparation

After EDA, other features were generated to capture the temporal and recent pollution patterns. Temporal data, including hour, day, and month, were collected and categorical data, including wind direction and monitoring station identification, were encoded for the machine-learning models. The historical information of PM2.5 was also added.

The target variable was simply the moving forward by 1 hour of PM2.5 concentration at each monitoring station to forecast the PM2.5 concentration for the next 1 hour. So, the forecasting relationship was set as:

$$X_t \rightarrow \mathrm{PM2.5}_{t+1}$$

where X_t is the information available at the current hour and $\mathrm{PM2.5}_{t+1}$ is the PM2.5 concentration of the next hour.

D. Chronological Train-Test Split

The chronological train-test split method was chosen because the forecasting of PM2.5 is a time-dependent problem, so random splitting is inappropriate. Approximately 80% of the data was used for training, and the remaining 20% was used for testing. The final cutoff time was 18:00 on May 12, 2016. All observations that occurred at the same time were partitioned together. The number of observations in the training set was 336,600, while the test set accounted for 84,156 observations. The time period separation made sure the models would be trained on past observations and tested on a time period they had never seen before, minimizing the likelihood of time period information leakage.

E. Model Development and Evaluation

A persistence baseline and four regression models—Linear Regression, Decision Tree, Random Forest, and XGBoost—were evaluated. In the persistence baseline, the concentration of PM2.5 for the next hour is assumed to be the same as the current hour:

$$\widehat{\mathrm{PM2.5}}_{t+1} = \mathrm{PM2.5}_t.$$

Models were assessed by Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2). The lower the MAE and RMSE was, the lower the prediction errors were, and the higher the R^2 was, the better the agreement between prediction and actual concentration.

F. Rolling Validation

A rolling validation strategy was implemented on the chosen model to further test temporal stability, with the number of windows increased to look at the stability over a larger temporal window. Three chronological validation folds were taken into account:

$$2013 \rightarrow 2014, \quad 2013\text{–}2014 \rightarrow 2015, \quad 2013\text{–}2015 \rightarrow 2016.$$

The model was retrained for each fold, with the historical period of the fold used to train the model and the next unseen year tested. Each fold's MAE, RMSE, and R^2 values were computed and the average for each were used to evaluate consistency of performance across time.

IV. Result & Discussion

The performance of the persistence baseline and four machine-learning models was assessed for the chronological test set using MAE, RMSE, and R^2. Table I shows the overall performance of the tested approaches.

A. Overall Model Performance

The persistence baseline achieved the lowest MAE of 9.99 μg/m^3, while Linear Regression obtained the lowest RMSE of 19.19 μg/m^3 and the highest R^2 of approximately 0.95. Among the machine-learning models, Linear Regression therefore demonstrated the strongest overall performance. Random Forest also performed competitively, whereas Decision Tree produced the highest prediction errors.

TABLE I
PERFORMANCE OF THE FORECASTING APPROACHES

Model	MAE (μg/m^3)	RMSE (μg/m^3)	R^2
Persistence Baseline	**9.99**	19.62	0.94
Linear Regression	10.15	**19.19**	**0.95**
Decision Tree	17.08	31.95	0.85
Random Forest	10.41	20.12	0.94
XGBoost	11.31	21.99	0.93

B. Comparative Performance Analysis

All the forecasting techniques were evaluated using the three performance measures as illustrated in Fig. 3. The results of Linear Regression and the persistence baseline were very close, suggesting that the current hour is already a large part of the information for predicting the next hour of PM2.5 concentration. Linear Regression, however, did a more robust job of minimizing larger prediction errors since it had a lower RMSE and higher R^2.

The comparison also indicates that the more complex models were not necessarily more accurate in predicting. Random Forest and XGBoost did not achieve a better performance than Linear Regression, and the single Decision Tree's performance was much worse. The results presented here suggest that simple models may also be competitive in the case of PM2.5 forecasting at 1 hour lead time.

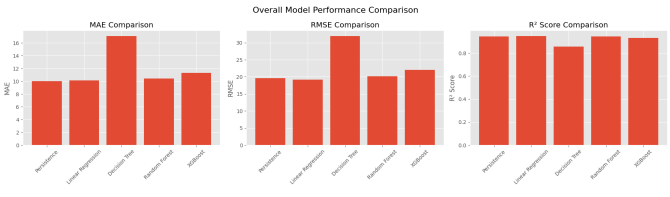


Fig. 3. Overall Model Performance Comparison.

C. Actual vs. Predicted PM2.5

The first 100 test observations were compared to the actual and predicted PM2.5 to visually compare the two. The overall trend of the concentrations and the changes in the concentrations (Fig. 4) generally agree with the Linear Regression model. But, for some rapid changes, the difference between the model and the observed PM2.5 is larger, either underestimating or overestimating the observed PM2.5.

In addition to the numerical results, the visual comparison shows that the fit of the overall short-term trend is quite good, although there are larger errors that occur around certain sudden fluctuations.

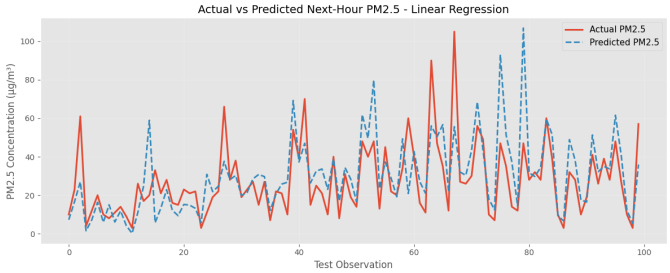


Fig. 4. Actual Vs Predicted Next-Hour PM2.5 - Linear Regression.

D. Rolling Validation Results

Linear Regression had relatively consistent results throughout the three validation periods. The average performance over the rolling validation set was MAE = 10.48 $\mu g/m^3$, RMSE = 19.40 $\mu g/m^3$, and R^2 = 0.9429, which were similar to the results from the main chronological test set. This implies that the model's performance did not necessarily have a significant impact on any one test period.

TABLE II
ROLLING VALIDATION RESULTS OF LINEAR REGRESSION

Training Period	Validation Year	MAE (μg/m^3) ↓	RMSE (μg/m^3) ↓	R^2 ↑
2013	2014	11.00	19.87	0.94
2013–2014	2015	10.61	19.77	0.94
2013–2015	2016	**9.83**	**18.54**	0.94
Average	—	**10.48**	**19.40**	**0.9429**

E. Discussion

The results of the experiments show that the more complex the model, the better the performance was not always. Ensemble models like Random Forest (RF), XGBoost, and LightGBM are reported to have high prediction accuracy in previous studies for PM2.5 levels [4], [8], [11], [14], [15], while temporal deep-learning architecture models such as attention-LSTM, CNN-LSTM, and Transformer-based models are reported to have high prediction accuracy for PM2.5 levels [5], [10], [11], [13]–[15]. However, the more complicated machine-learning models studied in this report were found to be more successful as a whole as compared to Linear Regression.

The other important finding is that the performance on the persistence baseline is quite good. The MAE is a little larger than the MAE of the Linear Regression, indicating that there is a lot of continuity in the PM2.5 concentration in the short term. Therefore, models with a high value of R^2 do not necessarily represent significantly better models than the straight line forecasting rules. It is more informative to compare machine-learning models with a meaningful baseline; such a comparison can give an insight into the usefulness of the predictions from the models.

It is not to claim that Linear Regression is better than other advanced models, but the accuracy of the forecast is based

on the data, features, forecast horizon, and the method of evaluation. Instead, the results suggest that simpler methods had a strong performance in the 1-hour-ahead forecasting framework explored in this study.

Conclusion

The PM2.5 forecasting study was performed in a time-aware machine-learning framework, with the Beijing Multi-Site Air Quality Dataset. The four machine learning models tested and compared for the metrics MAE, RMSE, R2 and a persistence baseline was computed. The machine-learning models outperformed the others in general with Linear Regression having the highest R^2 (around 0.95) and the lowest RMSE (19.19 $\mu g/m^3$) and the persistence baseline having the lowest MAE (9.99 $\mu g/m^3$). It was further noted, when the values were rolled within the lengths of time the values were validated over, that the Linear Regression was fairly consistent across the different lengths of time in which the values were validated.As can be seen from the results, the model performance of PM2.5 short-term prediction is not always better in terms of outcomes as the model complexity increases. The powerful persistence background also highlights the importance of contrasting the output of the machine-learning models with the performance of less complicated forecasting techniques. The framework can be extended to predict longer periods, incorporate more locations and environmental parameters, and incorporate more sophisticated temporal models like LSTM and Transformer-based models.

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